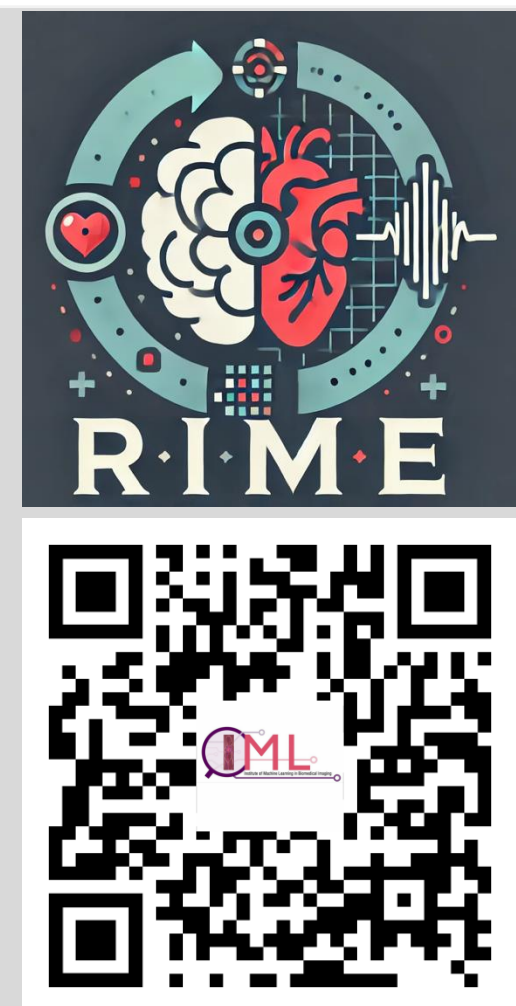


CUPA-T2*: Covariance-Aware Uncertainty Propagation and Alignment for T2* Mapping in Accelerated MRI

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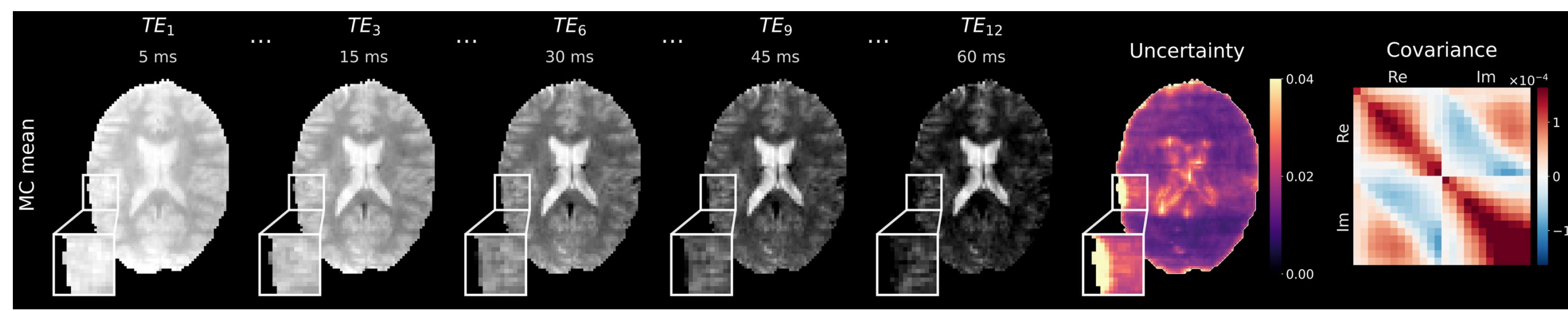


RIME
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Structured uncertainty matters for T2* estimation

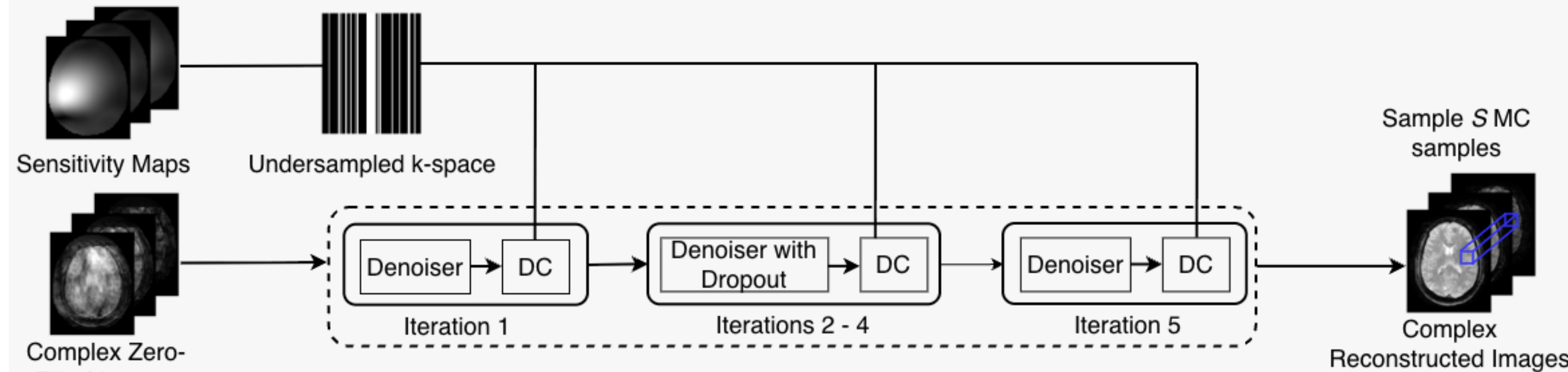
- Undersampling introduces reconstruction uncertainty, artifacts, and noise
- Uncertainty in multi-echo reconstruction are correlated across echoes
- Voxel-wise variance summaries ignore this structured reconstruction uncertainty
- Modeling inter-echo covariance may improve T2* fitting robustness



CUPA-T2* shows how structured reconstruction uncertainty can be leveraged to inform downstream T2* fitting

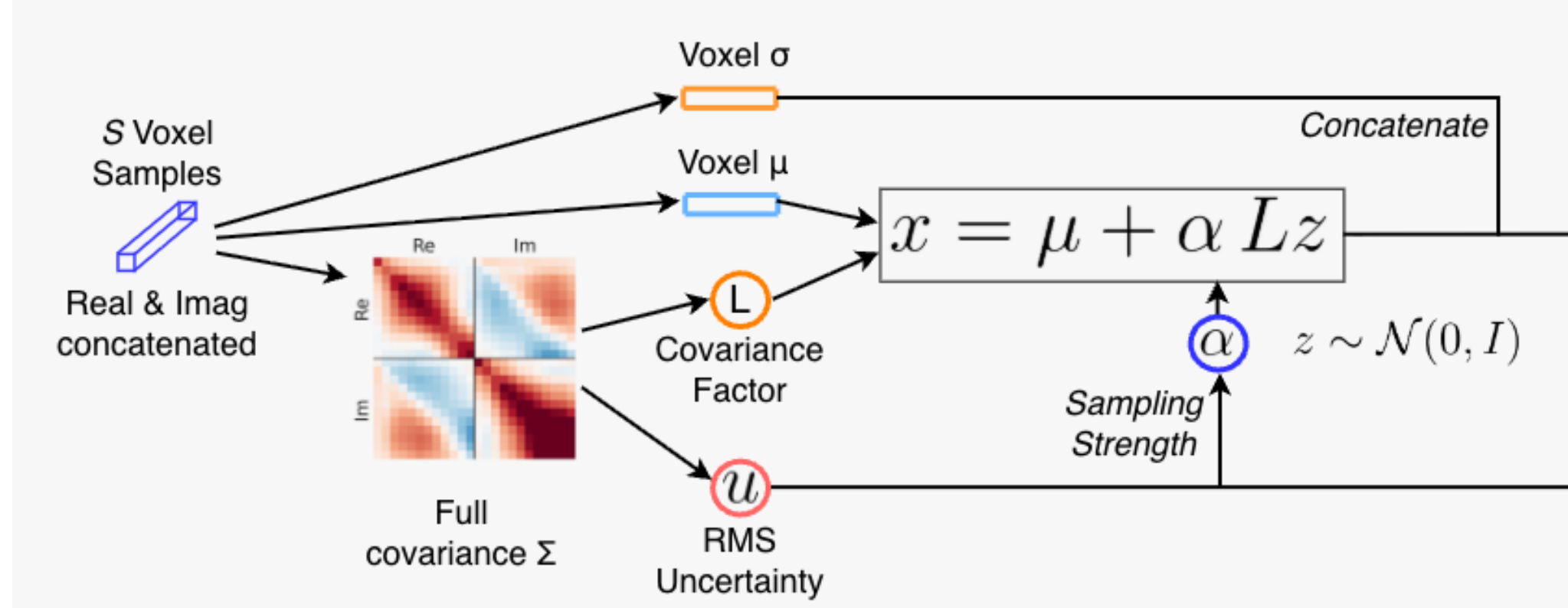
CUPA-T2*

A. Unrolled multi-echo reconstruction network with dropout

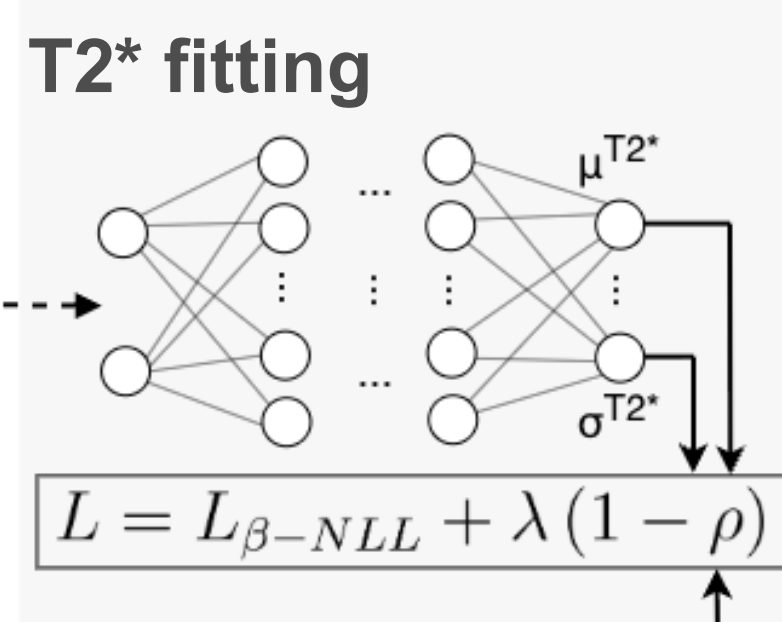


- After training, S=100 stochastic Monte-Carlo (MC) dropout reconstructions are generated per slice

B. Voxel-wise inter-echo covariance estimation



C. Heteroscedastic T2* fitting



- Inter-echo covariance is estimated from MC dropout reconstructions
- Inter-echo covariance is used for covariance-aware sampling
- Samples and per-echo uncertainty are fed into a heteroscedastic T2* regressor
- Predicted variance is aligned with reconstruction-derived uncertainty

Training and Evaluation

- Publicly available T2*-MOVE Dataset
- Trained a regular MLP (PUQ), Heteroscedastic MLP (Het), and CUPA-T2*
- T2* quality is assessed using normalized root mean square error (NRMSE) and Structural similarity index measure (SSIM)
- Aleatoric uncertainty is assessed using the calibration error (ECE), area under the risk-coverage curve (AURC), and Pearson correlation with reconstruction uncertainty (ρ)

Results

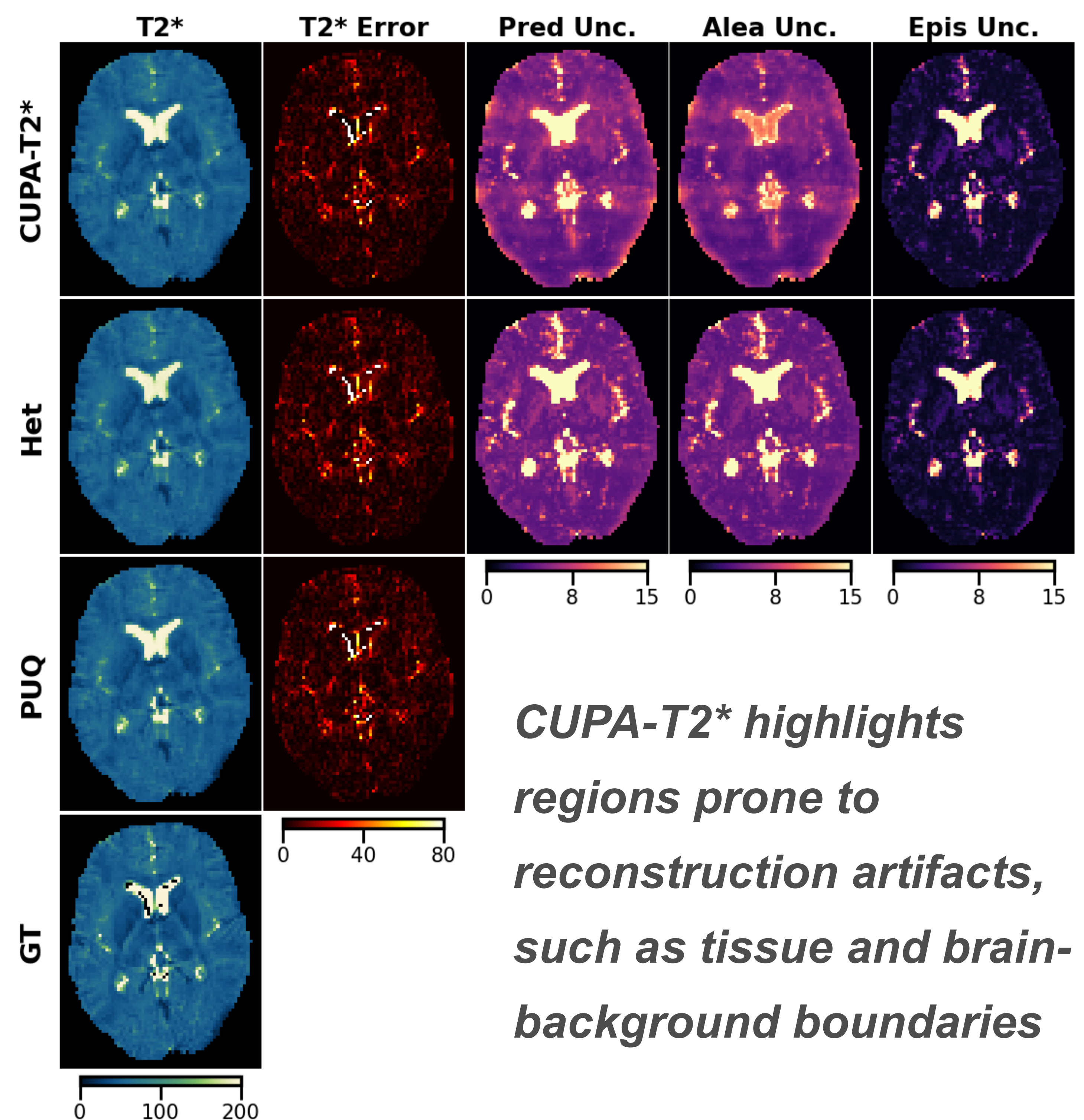
- Best metrics in **bold** and * indicating statistically superior to all others

Method	2×			3×			4×		
	WM ↓	GM ↓	CSF ↓	WM ↓	GM ↓	CSF ↓	WM ↓	GM ↓	CSF ↓
PUQ	0.100	0.117*	0.166*	0.137	0.168	0.250*	0.174	0.209	0.334*
Het	0.100	0.118	0.174	0.138	0.169	0.256	0.173	0.211	0.337
CUPA-T2*	0.099	0.119	0.171	0.135*	0.171	0.270	0.168*	0.210	0.359

- CUPA-T2* achieves lowest WM error at higher accelerations
- Performance is tissue-dependent

Method	2×			3×			4×		
	AURC ↓	ρ ↑	ECE ↓	AURC ↓	ρ ↑	ECE ↓	AURC ↓	ρ ↑	ECE ↓
Het	3.398*	0.280	0.031*	4.560*	0.259	0.033*	5.637*	0.189	0.036*
CUPA-T2*	3.471	0.766*	0.046	4.814	0.728*	0.051	5.910	0.651*	0.047

- The alignment between reconstruction and predicted uncertainty (ρ) is substantially improved by CUPA-T2*
- Improved alignment comes with a trade-off in ECE and AURC



CUPA-T2* highlights regions prone to reconstruction artifacts, such as tissue and brain-background boundaries